

Section 5.4 Exercises

Directions: For exercises 1 - 7, find the lengths of the missing sides of the right triangle if side a is opposite angle A , side b is opposite angle B , and side c is the hypotenuse.

1. $\cos B = \frac{4}{5}, a = 10$

2. $\sin B = \frac{1}{2}, a = 20$

3. $\tan A = \frac{5}{12}, b = 6$

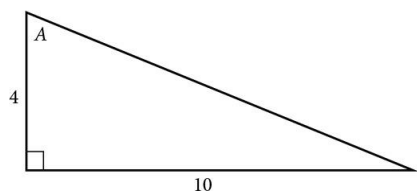
4. $\tan A = 100, b = 100$

5. $\sin B = \frac{1}{\sqrt{3}}, a = 2$

6. $a = 5, A = 60^\circ$

7. $c = 12, A = 45^\circ$

Directions: For exercises 8 - 13, use the right triangle below to evaluate the specified trigonometric function of angle A .



8. $\sin A$

9. $\cos A$

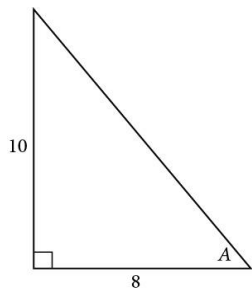
10. $\tan A$

11. $\csc A$

12. $\sec A$

13. $\cot A$

Directions: For exercises 14 - 19, use the given right triangle to evaluate the specified trigonometric function of angle A .



14. $\sin A$

15. $\cos A$

16. $\tan A$

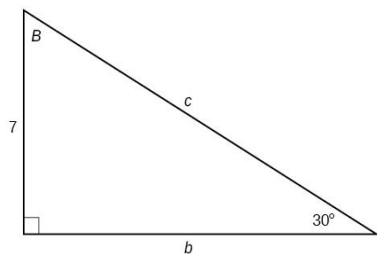
17. $\csc A$

18. $\sec A$

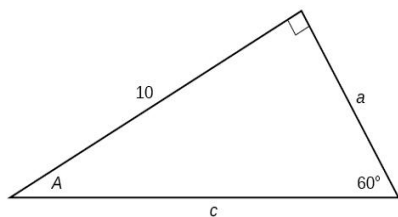
19. $\cot A$

Directions: For exercises 20 - 22, solve for the unknown sides of the given triangle.

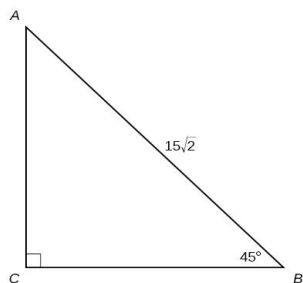
20.



21.

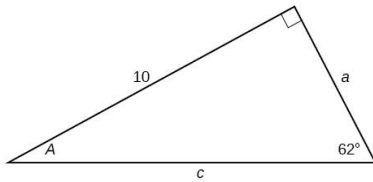


22.

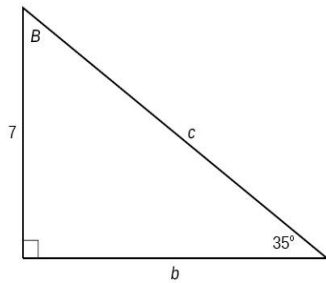


Directions: For exercises 23 - 32, use the given information from a right triangle to find the length of all unknown sides. Round all answers to four decimal places.

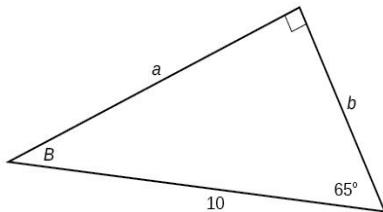
23.



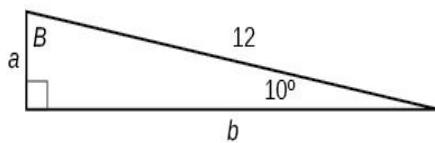
24.



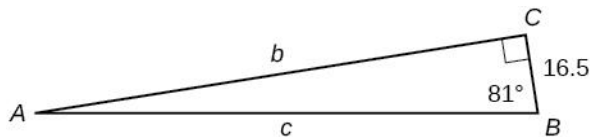
25.



26.



27.



28. $b = 15$, $B = 15^\circ$

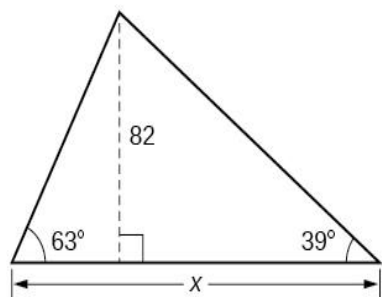
29. $c = 200$, $B = 5^\circ$

30. $c = 50$, $B = 21^\circ$

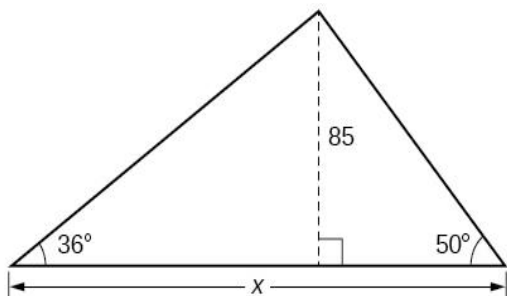
31. $a = 30$, $A = 27^\circ$

32. $b = 3.5$, $A = 78^\circ$

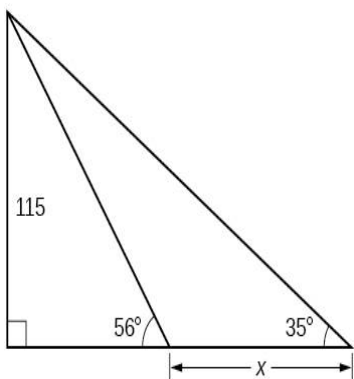
33. Find the value of x . Round the final answer to four decimal places.



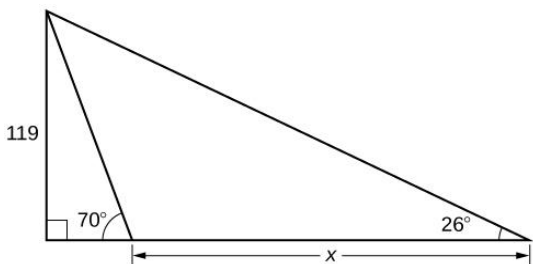
34. Find the value of x . Round the final answer to four decimal places.



35. Find the value of x . Round the final answer to four decimal places.



36. Find the value of x . Round the final answer to four decimal places.



37. A radio tower is located 400 feet from a building. From a window in the building, a person determines that the angle of elevation to the top of the tower is 36° , and that the angle of depression to the bottom of the tower is 23° . How tall is the tower? Round the final answer to four decimal places.

38. A radio tower is located 325 feet from a building. From a window in the building, a person determines that the angle of elevation to the top of the tower is 43° , and that the angle of depression to the bottom of the tower is 31° . How tall is the tower? Round the final answer to four decimal places.
39. A 200-foot tall monument is located in the distance. From a window in a building, a person determines that the angle of elevation to the top of the monument is 15° , and that the angle of depression to the bottom of the tower is 2° . How far is the person from the monument? Round the final answer to two decimal places.
40. A 400-foot tall monument is located in the distance. From a window in a building, a person determines that the angle of elevation to the top of the monument is 18° , and that the angle of depression to the bottom of the monument is 3° . How far is the person from the monument? Round the final answer to two decimal places.
41. There is an antenna on the top of a building. From a location 300 feet from the base of the building, the angle of elevation to the top of the building is measured to be 40° . From the same location, the angle of elevation to the top of the antenna is measured to be 43° . Find the height of the antenna. Round the final answer to two decimal places.
42. There is lightning rod on the top of a building. From a location 500 feet from the base of the building, the angle of elevation to the top of the building is measured to be 36° . From the same location, the angle of elevation to the top of the lightning rod is measured to be 38° . Find the height of the lightning rod. Round the final answer to two decimal places.
43. A 33-ft ladder leans against a building so that the angle between the ground and the ladder is 80° . How high does the ladder reach up the side of the building? Round the final answer to four decimal places.
44. A 23-ft ladder leans against a building so that the angle between the ground and the ladder is 80° . How high does the ladder reach up the side of the building? Round the final answer to four decimal places.
45. The angle of elevation to the top of a building in New York is found to be 9 degrees from the ground at a distance of 1 mile from the base of the building. Using this information, find the height of the building. Round the final answer to four decimal places.
46. The angle of elevation to the top of a building in Seattle is found to be 2 degrees from the ground at a distance of 2 miles from the base of the building. Using this information, find the height of the building. Round the final answer to four decimal places.
47. Assuming that a 370-foot tall giant redwood grows vertically, if I walk a certain distance from the tree and measure the angle of elevation to the top of the tree to be 60° , how far from the base of the tree am I? Round the final answer to four decimal places.